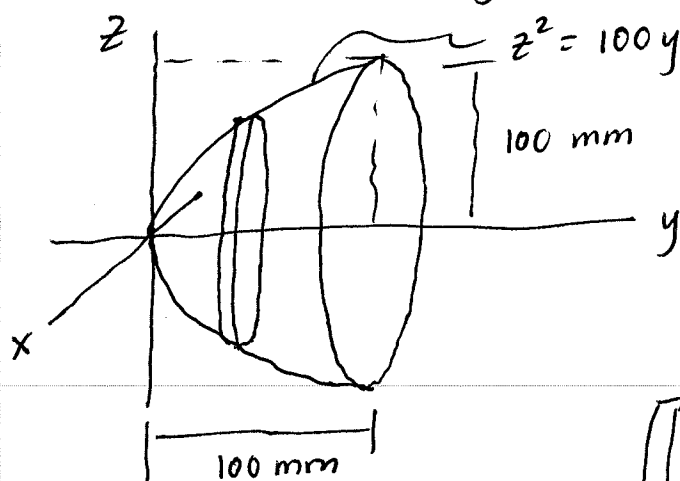


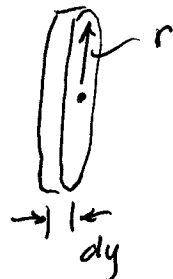
Ex Prob 9.7 $\bar{y}$ of a solid



Find $\bar{y}$.

$$\bar{y} = \frac{\int_V \tilde{y} dV}{\int_V dV}$$

$$\tilde{y} = y$$



$$dV = \pi r^2 dy$$

$$r^2 = z^2 = 100y$$

$$dV = \pi(100y) dy$$

$$\text{Volume} = \int_0^{100} (100\pi) y dy = (100\pi) \frac{1}{2} y^2 \Big|_0^{100}$$

$$\text{Vol} = \frac{100^3 \pi}{2} = \boxed{1.5708 \cdot 10^6 \text{ mm}^3 = \text{VOL}}$$

$$\bar{y} \text{ Numerator} = \int_0^{100} (100\pi) y \cdot y dy = (100\pi) \frac{1}{3} y^3 \Big|_0^{100}$$

$$= \frac{100^4 \pi}{3} =$$

$$\bar{y} = \frac{100 \cdot \frac{100^3 \pi}{3}}{\frac{100^3 \pi}{2}} = \frac{2 \cdot 100}{3} = \boxed{66.7 \text{ mm} = \bar{y}}$$